Cell Tuition
1-1 PRIVATE TUTORS

ALGEBRA

The Quadratic Formula

Name _____ Date _____ Calculator allowed Version 1

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KEY WORDS quadratic formula · coefficient *(tau whakarea)* · discriminant · solution *(otinga)*

A Naming a, b and c ●○○

Remember: Write the equation as $ax^2 + bx + c = 0$ first. Keep the sign with each number.

1. Write down the values of a, b and c.

a) $x^2 + 5x + 6 = 0$ _____ **b)** $2x^2 - 3x - 7 = 0$ _____

c) $x^2 - 9 = 0$ _____ **d)** $x^2 - 4x = 0$ _____

e) $5 - 2x - x^2 = 0$ _____ **f)** $3x^2 = 4x + 1$ _____

g) $x^2 + x = 12$ _____ **h)** $2x^2 = 5$ _____

B The discriminant ●●○

Remember: The discriminant is $b^2 - 4ac$. It is the part of the formula under the square root.

2. Work out $b^2 - 4ac$ for each equation.

a) $x^2 + 3x + 1 = 0$ **b)** $2x^2 - 3x - 5 = 0$ **c)** $x^2 - 6x + 9 = 0$

d) $3x^2 + 2x + 4 = 0$ **e)** $x^2 - 7 = 0$ **f)** $5x^2 - 4x - 2 = 0$

g) $-x^2 + 5x + 3 = 0$ **h)** $6x^2 + 7x + 2 = 0$ **i)** $2x^2 - x + 10 = 0$

C Solving with the formula ●●○

Remember: For $ax^2 + bx + c = 0$: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

3. Solve with the quadratic formula. These have exact answers.

a) $x^2 + 5x + 6 = 0$ **b)** $x^2 - 7x + 10 = 0$ **c)** $2x^2 + 5x + 2 = 0$

d) $x^2 + 2x - 8 = 0$ **e)** $2x^2 - 7x + 3 = 0$ **f)** $4x^2 + 4x + 1 = 0$

g) $x^2 - x - 12 = 0$ **h)** $4x^2 - 9 = 0$ **i)** $5x^2 + 9x - 2 = 0$

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D Answers to 2 decimal places ●●●

⚠ **Watch out:** Take care with negatives: if $b = -5$ then $-b = 5$ and $b^2 = 25$.

4. Solve with the quadratic formula. Give your answers to 2 decimal places.

a) $x^2 + 4x + 1 = 0$ **b)** $x^2 - 5x + 3 = 0$ **c)** $x^2 + 6x - 2 = 0$

d) $2x^2 + 3x - 4 = 0$ **e)** $3x^2 - 7x + 1 = 0$ **f)** $x^2 - 2x - 5 = 0$

g) $2x^2 - 8x + 3 = 0$ **h)** $5x^2 + 2x - 1 = 0$ **i)** $-x^2 + 4x + 3 = 0$

E How many solutions? ●●●

💡 **Remember:** Discriminant positive: two solutions. Zero: one solution. Negative: no solutions.

5. Use the discriminant to decide how many solutions each equation has.

a) $x^2 + 4x + 5 = 0$ **b)** $x^2 - 10x + 25 = 0$ **c)** $2x^2 + 7x + 1 = 0$

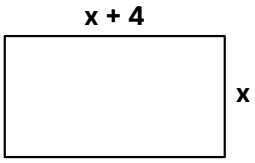
d) $3x^2 - 2x + 2 = 0$ **e)** $x^2 + x - 1 = 0$ **f)** $9x^2 + 6x + 1 = 0$

g) $2x^2 - 5x + 4 = 0$ **h)** $x^2 + 3 = 0$ **i)** $5x^2 + 3x - 2 = 0$

F Problems ●●●

6. A ball is thrown into the air. Its height h metres after t seconds is $h = -5t^2 + 12t + 2$. The ball lands when $h = 0$. After how many seconds does it land? Give your answer to 2 decimal places.

7. The rectangle has an area of 30 m². Write an equation and solve it to find x to 2 decimal places.



8. The square of a number plus the number itself equals 10. Find the two possible numbers, to 2 decimal places.

★ Extension

9. Solve $x^2 + 6x + 4 = 0$. Give your answers exactly, in the form $p \pm \sqrt{q}$.

10. The equation $x^2 + 6x + k = 0$ has exactly one solution. Find the value of k.

11. Solve $x(x + 3) = 5$. Give your answers to 2 decimal places.

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The Quadratic Formula

Name _____ Date _____ Calculator allowed Version 2

Scan to practise this topic online

KEY WORDS quadratic formula · coefficient *(tau whakarea)* · discriminant · solution *(otinga)*

A Naming a, b and c ●○○

Remember: Write the equation as $ax^2 + bx + c = 0$ first. Keep the sign with each number.

1. Write down the values of a, b and c.

a) $x^2 + 7x + 10 = 0$ _____ **b)** $3x^2 - 2x - 8 = 0$ _____

c) $x^2 - 16 = 0$ _____ **d)** $x^2 - 6x = 0$ _____

e) $7 - 3x - x^2 = 0$ _____ **f)** $2x^2 = 5x + 3$ _____

g) $x^2 + 2x = 15$ _____ **h)** $3x^2 = 7$ _____

B The discriminant ●●○

Remember: The discriminant is $b^2 - 4ac$. It is the part of the formula under the square root.

2. Work out $b^2 - 4ac$ for each equation.

a) $x^2 + 5x + 2 = 0$ **b)** $2x^2 - 5x - 3 = 0$ **c)** $x^2 - 8x + 16 = 0$

d) $2x^2 + 3x + 5 = 0$ **e)** $x^2 - 11 = 0$ **f)** $4x^2 - 6x - 1 = 0$

g) $-x^2 + 3x + 6 = 0$ **h)** $6x^2 + 11x + 3 = 0$ **i)** $3x^2 - x + 8 = 0$

C Solving with the formula ●●○

Remember: For $ax^2 + bx + c = 0$: $x = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

3. Solve with the quadratic formula. These have exact answers.

a) $x^2 + 7x + 12 = 0$ **b)** $x^2 - 8x + 15 = 0$ **c)** $2x^2 + 7x + 3 = 0$

d) $x^2 + 3x - 10 = 0$ **e)** $2x^2 - 9x + 4 = 0$ **f)** $4x^2 - 12x + 9 = 0$

g) $x^2 - x - 20 = 0$ **h)** $4x^2 - 25 = 0$ **i)** $5x^2 + 8x - 4 = 0$

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D Answers to 2 decimal places ●●●

⚠ **Watch out:** Take care with negatives: if $b = -5$ then $-b = 5$ and $b^2 = 25$.

4. Solve with the quadratic formula. Give your answers to 2 decimal places.

a) $x^2 + 6x + 2 = 0$ **b)** $x^2 - 7x + 5 = 0$ **c)** $x^2 + 4x - 3 = 0$

d) $2x^2 + 5x - 1 = 0$ **e)** $3x^2 - 8x + 2 = 0$ **f)** $x^2 - 4x - 6 = 0$

g) $2x^2 - 6x + 1 = 0$ **h)** $5x^2 + 3x - 1 = 0$ **i)** $-x^2 + 6x + 2 = 0$

E How many solutions? ●●●

Remember: Discriminant positive: two solutions. Zero: one solution. Negative: no solutions.

5. Use the discriminant to decide how many solutions each equation has.

a) $x^2 + 2x + 6 = 0$ **b)** $x^2 - 12x + 36 = 0$ **c)** $2x^2 + 9x + 3 = 0$

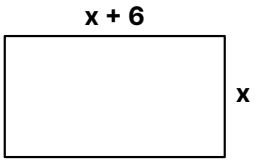
d) $4x^2 - 3x + 1 = 0$ **e)** $x^2 + 3x - 2 = 0$ **f)** $4x^2 + 20x + 25 = 0$

g) $3x^2 - 4x + 2 = 0$ **h)** $x^2 + 5 = 0$ **i)** $5x^2 + 2x - 3 = 0$

F Problems ●●●

6. A ball is thrown into the air. Its height h metres after t seconds is $h = -5t^2 + 15t + 4$. The ball lands when $h = 0$. After how many seconds does it land? Give your answer to 2 decimal places.

7. The rectangle has an area of 20 m². Write an equation and solve it to find x to 2 decimal places.



8. The square of a number minus 3 times the number equals 5. Find the two possible numbers, to 2 decimal places.

★ Extension

9. Solve $x^2 + 8x + 9 = 0$. Give your answers exactly, in the form $p \pm \sqrt{q}$.

10. The equation $x^2 - 10x + k = 0$ has exactly one solution. Find the value of k.

11. Solve $x(x - 2) = 7$. Give your answers to 2 decimal places.

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